Full-Stack Capstone

Eight build cycles, one architecture, one shipped container. React + Vite + TypeScript · Go + HTTP · persistent store · AI integration · auth · Docker.

6 COURSE 6 · FACILITATOR PACK

DURATION 8 hr (2 breaks + lunch)
AUDIENCE Capstone candidates
PREREQ Courses 1–4 + deployed tool + proposal

0:00–0:30	0:30–1:15	1:15–2:00	2:00–2:45	2:45–3:00	3:00–3:45	3:45–4:30	4:30–5:15	5:15–5:30	5:30–6:15	6:15–7:00	7:00–7:45	7:45–8:00
M1 Full-Stack Frontier FRAMING	**M2 Environment Setup** SETUP	**M3 Backend from Scratch** BUILD · ARCH 1/4	**M4 Data Layer** BUILD · ARCH 2/4	**Break** 15 MIN	**M5 Frontend from Scratch** BUILD · ARCH 3/4	**M6 Pages & Navigation** BUILD	**M7 AI Chat Integration** BUILD	**Break (often lunch)** 15 MIN	**M8 Auth & Middleware** BUILD	**M9 External Integrations** BUILD · *COMPRESS HERE*	**M10 Docker & Deploy** BUILD · ARCH 4/4	**Assessment & Wrap** CLOSE

PRE-SESSION PREP

Done 1 week before; verified at T-60 min

- **Capstone gate.** Every attendee submitted a proposal >= 7 days out. Reject anyone without an approved proposal — 8 hours is no place to discover scope drift.
- **Machines.** Each laptop runs Node 20+, Go 1.22+, Docker Desktop (admin), VS Code or equivalent, GenAI.mil access, GitHub or local repo. Smoke-test 24 h before.
- **Architecture diagram** printed and posted in the room (and on a side monitor). It is the one slide you return to nine times today.
- **Catering / break logistics.** Coffee at 0815, snacks at 1045 and 1530, lunch decision at 1230 (M7 break often becomes lunch). Hangry rooms break things.
- **Stretch goals doc.** Per-student: a 3-bullet stretch list to deploy on top of the base stack if they finish early.

WHEN THIS GOES SIDEWAYS

Top failure modes & recovery

- ☐ **Docker won't install (admin denied) on a student machine.**
  → Defer Docker to M10; pair the student with a Docker-enabled buddy for the deploy. They still ship.
- ☐ **External API in M9 returns 401s the entire module.**
  → Stub it. Hard-code a sample response. M9 is the compressible one for a reason.
- ☐ **A student wants to make the M5 table beautiful.**
  → Stop them. "Polish is for after 1700. Get data on the page." Bookmark their idea for the stretch list.
- ☐ **CORS / proxy errors stall multiple students at M5.**
  → Project the Vite proxy config slide. Walk through it once for the room. Do not re-explain individually — pair them up.
- ☐ **Behind 30+ minutes entering M8.**
  → Cut M9 entirely (announce it explicitly). Re-allocate to M10 so the container ships. Capture the missed module as Day-2 homework.

LIVE HAND-OFF CUES

Constant deck ↔ IDE switching — the architecture is the anchor

- **End of M2.** Return to the architecture slide. *Stop.* Point at it. "You will see this exact picture eight more times today."
- **Module-complete cadence.** Each build module ends with a 60-second "arch lights up" slide. Deck → diagram → back to IDE for the next build.
- **End of M3 (1st curl response).** Twenty seconds of silence. This is the emotional checkpoint of the day — let it land.
- **M9 compression rule.** If you're behind by 15 min entering M9, cut external integrations to a stub. Everyone ships a container by 1700 — no exceptions.
- **1700 wrap.** Each student demos their running container in 60 sec. You moderate; you do not debug.

POST-SESSION PREP

Issued in the 1745 wrap; capstone hand-off

- **Push the container** to your unit's registry within 72 h. URL goes in the program shared library.
- **Write the Unit Playbook** for hand-off to the next rotation: architecture diagram, run instructions, two known issues, one stretch goal. Use the Week 4 playbook template.
- **Run a 30-min teach-back** with two non-attendees in the next 30 days. Show the container. Show the architecture diagram. Take their first three questions to the program lead.
- **Capstone certificate** issued after playbook submission & teach-back evidence are received.

EXERCISE & ACTIVITY PROMPTS

Read these to the room verbatim

M1 (30 min). "You stop being a Power Apps builder today. Today you direct AI to build whatever the problem needs — backend, frontend, data, auth. Look at the architecture. Four layers. Eight builds. One container at 1700."

M2 (45 min). "Most boring module of the day; most expensive if we skip it. Verify Node, Go, Docker. AI walks each of you through your own machine — do not let me do it for you."

M3 (45 min). "Direct AI to write a Go HTTP server that returns a hard-coded JSON array on `GET /items`. You curl it. When you see `200 OK`, raise a hand. Twenty-second pause for everyone to feel that."

M4 (45 min). "Interface-first. Define what the data layer *does* before you decide how it stores. AI implements the interface against an in-memory store first; persistence in a stretch."

M5–M6 (90 min). "React + Vite + TypeScript. Proxy to the Go backend. Render the items table. Then add routes for two pages and a nav. Resist beauty — we add features in M6, polish never."

M7 (45 min). "Wire the OpenAI-compatible chat endpoint to your backend. Stream tokens to the React UI. The first time it streams, applaud once."

M8 (45 min). "Auth middleware on the backend; login route on the frontend. Single test user. Real password hashing. No production claims yet."

M9 (45 min, compressible). "One external integration of your choice from the proposal. If the API gives you trouble — stub it. M10 is non-negotiable."

M10 (45 min). "Dockerfile, build, run. Ship the container. URL or screenshot in chat by 1745."

> **ANCHOR PHRASE** "Architecture first. Light up one box at a time. Everyone ships a container by 1700."